

Passages - Machine Learning for Asset Managers

You Need a Theory, backtesting is not a research tool. Backtests can never prove that a strategy is a true positive, and they may only provide evidence that a strategy is a false positive. Never develop a strategy solely through" backtests. Strategies must be supported by theory, not historical simulations. Your theories must be general enough to explain particular cases, even if those cases are black swans.

The most insightful use of ML in finance is for discovering theories. You may use ML successfully for making financial forecasts; however, that is not necessarily the best scientific use of this technology (particularly if your goal is to develop high-capacity investment strategies).

Imagine if physicists had to produce theories in a universe where the fundamental laws of nature are in a constant flux; where publications have an impact on the very phenomenon under study; where experimentation is virtually impossible; where data are costly, the signal is dim, and the system under study is incredibly complex ... I feel utmost admiration for how much financial academics have achieved in the face of paramount adversity.

Bet sizing is at least as important as price prediction, and portfolio construction is arguably even more important.

Selection bias comes from fitting the model to perform well on the test set, not the train set.

A poorly performing backtest is an opportunity to fix the research process, not an opportunity to fix a particular investment strategy.

luck out.

No backtest can replace a theory, for at least two reasons: (1) backtests cannot simulate black swans - only theories have the breadth and depth needed to consider the never-before-seen occurrences; (2) backtests may insinuate that a strategy is profitable, but they do not tell us why.

They are not a controlled experiment. Only a theory can state the cause effect mechanism, and formulate a wide range of predictions and implications that can be independently tested for facts and counter facts. contains an empirical analysis, where the methods explained are put to the test in Monte Carlo experiments.

The first step in building a theory is to collect data that illustrate how some variables relate to each other. In financial settings, those data often take the form of a covariance matrix. We use covariance matrices to run regressions, optimize portfolios, manage risks, search for linkages, etc.

Many ML applications in finance do not require any historical data at all. Examples include risk analysis, portfolio construction, outlier detection, feature importance, and bet-sizing methods.

Financial ML applications, like sentiment analysis, deep hedging, credit ratings, execution, and private commercial data sets, enjoy an abundance of data.

The most successful quantitative

investment firms in history rely primarily on ML, not econometrics, and the current predominance of econometrics in graduate studies prepares students for academic careers, not for jobs in the industry.

This does not mean that econometrics has outlived its usability. ML and econometrics complement each other, because they have different strengths. For example, ML can be particularly helpful at suggesting to researchers the ingredients of a theory and econometrics can be useful at testing a theory that is well grounded on empirical observation.

While shrinkage is somewhat helpful in absence of denoising, it adds no benefit in combination with denoising. This is because shrinkage dilutes the noise at the expense of diluting some of the signal as well.

The Marcenko-Pastur theorem gives us the distribution of the eigenvalues associated with a random matrix. By fitting this distribution, we can discriminate between eigenvalues associated with signal and eigenvalues associated with noise.

Detoning is the principal components analysis analogue to computing beta-adjusted (or market-adjusted) returns in regression analysis.

A detoned correlation matrix cannot be used directly for mean-variance portfolio optimization. Instead, we can optimize a portfolio on the selected (non-zero principal components, and map the optimal allocations f^* back to the original basis. The optimal allocations in the original basis are

$$\omega^* = W_+ f^*,$$

where W_+ contains only the eigenvectors that survived the detoning process (i.e., with a nonnull eigenvalue), and f^* is the vector of optimal allocations to those same components.

In an efficient market all securities have the same expected Sharpe ratio.

The correlation matrix's condition number is the ratio between its maximal and minimal (by moduli) eigenvalues.

The key idea behind entropy is to quantify the amount of uncertainty associated with a random variable. Information theory is also essential to ML, because the primary goal of many ML algorithms is to reduce the amount of uncertainty involved in the solution to a problem.